

Analysis 1 HW 2

5

Definitions

A is a nonempty set of real numbers bounded below.

$$-A = \{-x \mid x \in A\}.$$

To prove: $\inf(A) = -\sup(-A)$.

Proof. Since A is nonempty and bounded below, $y = \inf(A)$ exists. It follows that

$$\begin{aligned} y &\leq x \quad \forall x \in A \\ \implies -x &\leq -y \quad \forall x \in A \\ \implies -x &\leq -y \quad \forall -x \in -A \end{aligned}$$

$\therefore -y$ is an upper bound of $-A$.

Also,

$$\begin{aligned} \forall x > y \quad \exists x' \in A \text{ such that } x' < x \\ \implies \forall -x < -y \quad \exists -x' \in -A \text{ such that } -x' > -x \end{aligned}$$

$\therefore -y$ is the smallest upper bound of $-A$.

$$-\inf(A) = \sup(-A).$$

□

6

Definitions

$b \in \mathbb{R}, b > 1$.

$m, n, p, q \in \mathbb{Z}, n > 0, q > 0$.

$r \in \mathbb{Q}, r = m/n = p/q$.

$a^{1/n}$ is defined as the positive solution to the equation $y^n = a$ (Rudin, 1.21).

Lemma 6.1. $(a_1 a_2 \cdots a_k)^{1/n} = a_1^{1/n} a_2^{1/n} \cdots a_k^{1/n}$

Proof. From the corollary of 1.21, we know that $(a_1 a_2)^{1/n} = a_1^{1/n} a_2^{1/n}$. Assume the result holds for some $k \geq 2$. Then, $(a_1 a_2 \cdots a_{k+1})^{1/n} = (a_1 a_2 \cdots a_k)^{1/n} a_{k+1}^{1/n} = a_1^{1/n} a_2^{1/n} \cdots a_{k+1}^{1/n}$. By the principle of mathematical induction, the lemma is true for all $k \geq 2$. □

Lemma 6.2. $(b^m)^{1/n} = (b^{1/n})^m$

Proof. If $m \geq 0$,

$$(b^m)^{1/n} = \left(\prod_{i=1}^m b \right)^{1/n} = \prod_{i=1}^m b^{1/n} = (b^{1/n})^m$$

from 6.1. A similar argument works for $m < 0$. □

(a) **To prove:** $(b^m)^{1/n} = (b^p)^{1/q}$.

Proof. We know that $mq = pn$. Thus, $b^{mq} = b^{pn}$ or $(b^m)^q = (b^p)^n$. It follows that

$$\begin{aligned} ((b^m)^q)^{1/n} &= b^p && \text{Rudin, 1.21} \\ \implies ((b^m)^{1/n})^q &= b^p && 6.2 \\ \implies (b^m)^{1/n} &= (b^p)^{1/q} && \text{Rudin, 1.21} \end{aligned}$$

□

Definition 6.3. $b^r \equiv (b^m)^{1/n}$.

(b) **To prove:** $b^{r+s} = b^r b^s$, $r, s \in \mathbb{Q}$.

Proof. Let $s = c/d$, $c, d \in \mathbb{Z}$, $d > 0$. Then,

$$\begin{aligned} b^{r+s} &= (b^{md+cn})^{1/nd} && 6.3 \\ &= (b^{md} b^{cn})^{1/nd} \\ &= (b^{md})^{1/nd} \cdot (b^{cn})^{1/nd} && 6.1 \\ &= (b^m)^{1/n} \cdot (b^c)^{1/d} && 6(a) \\ &= b^r b^s && 6.3 \end{aligned}$$

□

(c)

Lemma 6.4. If $x, y, z, w \in \mathbb{R}^+$, $x \leq y$, $z \leq w$, then $xz \leq yw$.

Proof. If $x < y$ and $z < w$,

$$\begin{aligned} x < y &\implies zx < zy && \text{Rudin, 1.18(b)} \\ z < w &\implies yz < yw && \text{Rudin, 1.18(b)} \\ \therefore xz &< yw && \text{Rudin, 1.5(ii)} \end{aligned}$$

If $x = y$ and $z < w$, $xz < yw$ from Rudin, 1.18(b).

If $x < y$ and $z = w$, $xz < yw$ from Rudin, 1.18(b).

If $x = y$ and $z = w$, $xz = yw$. □

Lemma 6.5. If $r, s \in \mathbb{Q}$, $r > s$ and $b \in \mathbb{R}$, $b > 1$ then $b^r > b^s$.

Proof. Let $s = c/d$, $c, d \in \mathbb{Z}$, $d > 0$. Then, $md > cn$, which implies $b^{md} > b^{cn}$. For the sake of contradiction, assume $(b^{md})^{1/nd} \leq (b^{cn})^{1/nd}$. Then,

$$\begin{aligned} \prod_{i=1}^{nd} (b^{md})^{1/nd} &\leq \prod_{i=1}^{nd} (b^{cn})^{1/nd} & 6.4 \\ \implies ((b^{md})^{nd})^{1/nd} &\leq ((b^{cn})^{nd})^{1/nd} & 6.1 \\ \implies b^{md} &\leq b^{cn} & \text{Rudin, 1.21} \end{aligned}$$

which contradicts $b^{md} > b^{cn}$. Thus,

$$\begin{aligned} (b^{md})^{1/nd} &> (b^{cn})^{1/nd} \\ \implies (b^m)^{1/n} &> (b^c)^{1/d} & 6(a) \\ \implies b^r &> b^s & 6.3 \end{aligned}$$

□

This lemma can be extended to say that if $r \geq s$ then $b^r \geq b^s$, since $r = s$ implies $b^r = b^s$.

Definition 6.6. $B(x) \equiv \{b^t \mid t \in \mathbb{Q}, t \leq x\}$, $x \in \mathbb{R}$.

To prove: $b^r = \sup(B(r))$, $r \in \mathbb{Q}$.

Proof. Since r is rational, $r - 1$ is rational. $r - 1 < r$, so $b^{r-1} \in B(r)$. Hence, $B(r)$ is nonempty.

For all $b^t \in B(r)$ we have $t \leq r$, which implies $b^t \leq b^r$ (6.5). Hence, b^r is an upper bound of $B(r)$ (which implies $\sup(B(r))$ exists).

Also, since $r \in \mathbb{R}$ and $r \leq r$, $b^r \in B(r)$. Thus, no $b^t < b^r$ can be an upper bound of $B(r)$.

$\therefore \sup(B(r)) = b^r$. □

Definition 6.7. $b^x \equiv \sup B(x)$, $x \in \mathbb{R}$.

(d)

Lemma 6.8. For all $x, y \in \mathbb{R}^+$ and $m, n \in \mathbb{Z}$, $m \geq 1$, $x^m < y^n$ implies $x < (y^n)^{1/m}$.

Proof. It suffices to prove that $x \geq (y^n)^{1/m} \implies x^m \geq y^n$.

$$\begin{aligned} x &\geq (y^n)^{1/m} \\ \implies x^m &\geq ((y^n)^{1/m})^m & 6.4 \\ \implies x^m &\geq y^n & \text{Rudin, 1.21} \end{aligned}$$

□

Lemma 6.9. For all $b > 1$, the set $\{[b^n, b^{n+1}) \mid n \in \mathbb{Z}\}$ is a partition of $\mathbb{R}^+$.

I have not been able to prove this rigourously.

Lemma 6.10. For all $\alpha, \beta \in \mathbb{R}^+$, $\alpha < \beta$ there exists a number of the form b^r where $b \in \mathbb{R}$, $b > 1$, $r \in \mathbb{Q}$ such that $\alpha < b^r < \beta$.

Proof. Since $\alpha < \beta$, $\beta/\alpha > 1$. From 6.9, there exists $k \in \mathbb{Z}$ such that $b \in [(\beta/\alpha)^{k-1}, (\beta/\alpha)^k)$. Thus, $b < (\beta/\alpha)^k$, or $\alpha^k b < \beta^k$. Since $b > 1$, we have $k - 1 \geq 0$ or $k \geq 1$ (required for Lemma 6.8).

Also from 6.9, there exists $l \in \mathbb{Z}$ such that $b^{l-1} \leq \alpha^k < b^l$. Multiplying this inequality by b yields $b^l \leq \alpha^k b$.

Chaining the above inequalities gives

$$\begin{aligned} b^{l-1} &\leq \alpha^k < b^l \leq \alpha^k b < \beta^k \\ \implies \alpha^k &< b^l < \beta^k \\ \implies \alpha &< b^{l/k} < \beta. \end{aligned} \tag{6.8}$$

□

To prove: $b^{x+y} = b^x b^y \forall x, y \in \mathbb{R}$.

Proof. Let $x + y \notin \mathbb{Q}$. This allows us to ignore the slackness of the inequality in Definition 6.6. Let $x_1, x_2 \in \mathbb{Q}$ such that $x_1 < x < x_2$ (the existence of such x_1 and x_2 is guaranteed by the archimedean property). Similarly, let $y_1, y_2 \in \mathbb{Q}$ such that $y_1 < y < y_2$. Note that adding the two inequalities gives $x_1 + y_1 < x + y < x_2 + y_2$.

Now, since $x_1 < x$, it must be that $b^{x_1} \in B(x)$ by Definition 6.6. So, $b^{x_1} \leq \sup B(x)$, which implies $b^{x_1} \leq b^x$ by Definition 6.7. Since $x_1 \neq x$, we have $b^{x_1} < b^x$. From 6.5, we have $b^{x_1} < b^{x_2}$. It cannot be the case that $b^{x_2} < b^x$, since $x_2 \notin B(x)$. Thus, we have $b^{x_1} < b^x < b^{x_2}$. Similarly, $b^{y_1} < b^y < b^{y_2}$.

$$\begin{aligned} b^{x_1} &< b^x < b^{x_2} \text{ and } b^{y_1} < b^y < b^{y_2} \\ \implies b^{x_1} b^{y_1} &< b^x b^y < b^{x_2} b^{y_2} & 6.4 \\ \implies b^{x_1+y_1} &< b^x b^y < b^{x_2+y_2} & 6(b) \end{aligned}$$

This tells us that $b^x b^y$ is an upper bound for the set $S_1 \equiv \{b^t \mid t \in \mathbb{Q}, t < x + y\}$ and a lower bound for the set $S_2 \equiv \{b^t \mid t \in \mathbb{Q}, t > x + y\}$. Thus, it must be the case that

$$\sup S_1 \leq b^x b^y \leq \inf S_2.$$

$\sup S_1$, from Definition 6.7, is b^{x+y} . Thus, it suffices to prove that $\sup S_1 = \inf S_2$. FTSOC, assume otherwise.

(i) $\sup S_1 < \inf S_2$

From 6.10, there exists b^r , $r \in \mathbb{Q}$ such that $\sup S_1 < b^r < \inf S_2$. $b^r \notin S_1$, which implies that $r \geq x + y$, which is equivalent to $r > x + y$ since $x + y$ is not rational. Similarly, $b^r < \inf S_2$ implies that $b^r \notin S_2$, and $r \leq x + y$ which is equivalent to $r < x + y$. $\Rightarrow \Leftarrow$

(ii) $\inf S_2 < \sup S_1$

From the definition of supremum, there must exist $b^r \in S_1$ such that $\inf S_2 < b^r \leq \sup S_1$. It follows that $r < x + y$.

Similarly, from the definition of infimum, there must exist $b^{r'} \in S_2$ such that $\inf S_2 \leq b^{r'} < b^r$. It follows that $r' > x + y$.

$$r' > x + y > r \implies r' > r.$$

$$\text{However, } b^{r'} < b^r \implies r' < r. \quad \implies \Leftarrow$$

Thus, $b^{x+y} = \sup S_1 = \inf S_2 = b^x b^y$.

If $x + y \in \mathbb{Q}$, it suffices to prove that $\sup\{b^t \mid t \in \mathbb{Q}, t < x + y\} = \sup\{b^t \mid t \in \mathbb{Q}, t \leq x + y\} = b^{x+y}$. This is trivially true from Lemma 6.10 ($\forall c < b^{x+y} \exists b^r$ such that $c < b^r < b^{x+y}$. b^r is clearly a member of the first set. Hence c is not an upper bound). $\square$

7

Definitions

$b \in \mathbb{Q}, b > 1$.

$y \in \mathbb{Q}, y > 0$.

(a) **To prove:** $b^n - 1 \geq n(b - 1) \forall n \in \mathbb{Z}^+$.

Proof. This can be proved using mathematical induction. For $n = 1$, $b - 1 \geq b - 1$. Let the inequality hold for some $k \in \mathbb{Z}^+$. Then,

$$\begin{aligned} b^k - 1 &\geq k(b - 1) \\ \implies b(b^k - 1) &\geq k(b - 1) \\ \implies b^{k+1} - 1 &\geq k(b - 1) + b - 1 \\ \implies b^{k+1} - 1 &\geq (k + 1)(b - 1) \end{aligned}$$

$\square$

(b) **To prove:** $b - 1 \geq n(b^{1/n} - 1) \forall n \in \mathbb{Z}^+$.

Plug $b^{1/n}$ in place of b in 7(a).

(c) **To prove:** $t > 1$ and $n > (b - 1)/(t - 1) \implies b^{1/n} < t$.

Proof. From (b), we have

$$\frac{b - 1}{n} + 1 \geq b^{1/n}.$$

Rearranging the given inequality gives us

$$t > \frac{b - 1}{n} + 1.$$

It follows that $t > b^{1/n}$. $\square$

(d) **To prove:** $b^w < y \implies (\exists n \in \mathbb{Z}^+ \text{ such that } b^{w+1/n} < y).$

Proof. Let $t = y/b^w$. $t > 1$. Consider $n \in \mathbb{Z}^+$ such that $n > (b-1)/(t-1)$. The existence of such an n is guaranteed by the archimedean property. Then, by 7(c), we have $b^{1/n} < y/b^w$, which gives us $b^{w+1/n} < y$. $\square$

(e) **To prove:** $b^w > y \implies (\exists n \in \mathbb{Z}^+ \text{ such that } b^{w-1/n} > y).$

Proof. Let $t = b^w/y$. $t > 1$. Consider $n \in \mathbb{Z}^+$ such that $n > (b-1)/(t-1)$. The existence of such an n is guaranteed by the archimedean property. Then, by 7(c), we have $b^{1/n} < b^w/y$, which gives us $y < b^{w-1/n}$. $\square$

(f) $A \equiv \{w \mid b^w < y\}$

$x = \sup A$

To prove: $b^x = y$.

Proof. FTSOC, assume otherwise.

(i) $b^x < y$

From 7(d), there exists $n \in \mathbb{Z}^+$ such that $b^{x+1/n} < y$, which implies $x + 1/n \in A$. However, it is given that x is an upper bound of A . $\Rightarrow \Leftarrow$

(ii) $b^x > y$

From 7(e), there exists $n \in \mathbb{Z}^+$ such that $b^{x-1/n} > y$. Clearly, $x - 1/n$ is an upper bound of A . However, it is given that x is the smallest upper bound of A . $\Rightarrow \Leftarrow$

$\square$

(g) **To prove:** The x obtained in (f) is unique.

x is defined as the supremum of A in (f). The supremum of a set is always unique.

9

Definitions

$$z = a + bi$$

$$w = c + di$$

$$<: \mathbb{C} \rightarrow \mathbb{C} : z < w \iff (a < c) \text{ or } (a = c \text{ and } b < d).$$

To show that this order turns $\mathbb{C}$ into an ordered set, we need to show trichotomy and transitivity. The former is easily shown as follows.

(i) $a < c \implies z < w$

(ii) $a = c$ and

(a) $d < b \implies z < w$

$$(b) \quad d = b \implies z = w$$

$$(c) \quad d > b \implies w < z$$

$$(iii) \quad a > c \implies w < z$$

To show transitivity, consider $z, w, x (= e + fi) \in \mathbb{C}$, such that $z < w$ and $w < x$.

$$(a < c \text{ or } (a = c \text{ and } b < d)) \text{ and } (c < e \text{ or } (c = e \text{ and } d < f))$$

$$\begin{aligned} \implies & (a < c \text{ and } c < e) \text{ or} \\ & (a < c \text{ and } (c = e \text{ and } d < f)) \text{ or} \\ & ((a = c \text{ and } b < d) \text{ and } c < e) \text{ or} \\ & ((a = c \text{ and } b < d) \text{ and } (c = e \text{ and } d < f)) \end{aligned}$$

$$\begin{aligned} \implies & (a < e) \text{ or} \\ & (a < e \text{ and } d < f) \text{ or} \\ & (a < e \text{ and } b < d) \text{ or} \\ & (a = e \text{ and } b < f) \end{aligned}$$

$$\begin{aligned} \implies & (a < e) \text{ or} \\ & (a = e \text{ and } b < f) \end{aligned}$$

$$\implies z < x$$

Complex numbers under lexicographic order have the smallest upper bound property. consider the set $S = \{a + ib | a \in A, b \in B\}$ where A and B are sets of real numbers. If $x = \sup A$ and $y = \sup B$, then $x + iy$ is clearly an upper bound of S . For any $z \in S$, $z < x + iy$, z' can be found such that $z < z' < x + iy$ thanks to the smallest upper bound property of the real numbers.